Diabetes Predicting mHealth Application Using Machine Learning

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***Abstract*— With the advancement of information technologies, mobile health (mHealth) technologies can be leveraged for patient self-management, patient diagnosis and determining the probability of being affected by some disease. Diabetes mellitus is a chronic and lifestyle disease and millions of people from all over the world fall victim to it. Although there are some mobile apps keeping track of calories, sugar taken, medicine doses, lifestyle, blood glucose, blood pressure, weight of individuals and giving suggestion about food, exercises to prevent or control diabetes, no application has been found that was explicitly developed to analyze the risk of being a diabetic patient. Therefore, the objective of this paper is to develop an intelligent mHealth application based on machine learning to assess his/her possibility of being diabetic, prediabetic or nondiabetic without the assistance of any doctor or medical tests.**

***Keywords*—mobile health, diabetes, naive Bayes classifier, survey, machine learning, prediction application.**

I. Introduction

In Mobile Health (mHealth) includes wireless technologies to provide various data contents and health related services through devices such as smart phones, tablets, PCs etc [1]. World Health Organization (WHO) in 2011 has found that over 83% of member states reported implementation of at least four or more mHealth solutions [2]. By 2015, number of apps concerning health issues increased to 165,000 [3].

According to WHO [4], by 2012 an estimated 422 million people were living with diabetes and by 2030, this number is estimated to increase to about 552 million [5, 6]. Even in Bangladesh about 8.4 million of the adult population are carrying this deadly disease [7] and nearly 51.2% of the population are unaware of it [8]. So, globally diabetes has become a primary health concern.

Among many other initiatives, mHealth service for diabetic patients is very popular. A systematic review conducted by Ahn [9] found a total 656 diabetes related apps including both iOS and Android app. Basar et al [10] filtered 43 diabetes related apps.

There are mainly three forms of diabetes, type 1, type 2 and gestational diabetes. According to Raha [11], 5% of total people suffering from diabetes have type 1 diabetes (juvenile diabetes) but the possibility of being affected by this kind of diabetes is not predictable [11]. Gestational diabetes is diagnosed during pregnancy. The most common form is the type 2 diabetes; the symptoms of which can be predicted and this can be prevented through proper care and knowledge. A prediction system depicted in study [12] takes into count 23 features and applies regression technique to predict if a person can be diabetic and at what age. Investigation of the performances of different machine learning approaches for predicting diabetes from medical records is presented in [13].

The alarming increase in number of diabetes patients, deaths caused by diabetes and the need of an application that requires the most basic features for being used by any individual to predict possibilities of being diabetic were the main motivations behind our work. The objective of this paper is to develop an intelligent predictive mHealth application to predetermine individual's possibility of being *diabetic*, *prediabetic* or *nondiabetic*; for this prediction individual won't need any medical assistance. In order to attain the objective, we have reviewed the existing related research works, carried out a survey study using factors that are easily known to general people, analyzed the collected data, used an algorithm based on machine learning technique and developed an mobile application using that algorithm. The features chosen was kept very simple so that any individual can easily

Later sections of this paper are arranged as follows. Section II gives idea about related research works. The research methodology has been explained briefly in Section III. The implementation of the diabetes prediction application including the prediction algorithm, its evaluation and development process of the mHealth app has been discussed in Section IV. Finally the discussion and conclusion are stated in Section V.

II. Literature Review

In this section, we have discussed related research works focusing on diabetes based mHealth technologies and the prediction systems or algorithms for different diseases.

Till now, several studies have been conducted to highlight the importance of mHealth technology in health care and self-management [14]. A systematic review that has been conducted by Blaya et al [15] found that mobile phones improved efficiency of health services immensely and expanded the scope of delivering treatment to thousands of patients in an effective manner. FTA (Few Touch Application) tool, which is a combination of 10-app based systems is described in [16].The prime focus of a study conducted in [17] is diabetes management and monitoring. They have presented a METABO that records dietary, physical activity, medication and medical information. In [18] an application is developed for personalized health care of diabetic patients that provides decision on drug doses for maintaining stable glucose level.

Many other studies have been carried out on the usefulness of mHealth apps available in the market. In [19], a centralized resource is developed which contains information about more than 60,000 mHealth apps from both Google Play and Apple app store. In another study, Comstock [20] have predicted that by 2018, there will be 24 million diabetes app users. Table I provides a summary of few most downloaded apps with greater positive review which are available in Google Play Store [21].

TABLE I: Diabetes Related Mobile Applications

Name	Features
1. DiabetesM	• Keeps track of every aspect of diabetes treatment. • Generates reports, graphs and statistics based on stored records. • Reports can be further sent as emails to physicians.
2. Glucose buddy	• Record user's blood glucose level, food intake, blood pressure, medication etc. • Provides reminder for blood sugar testing after certain intervals
3. Diabetes Monitor	• Easy and intuitive recording, analyzing and better control of diabetes. • Measures blood sugar level and other related factors. • Deals with the activities that have impact on blood sugar.
4. Diabetes Predict Prevent	• Assesses the likelihood of developing diabetes by a 3 minute questionnaire. • On a positive result of the test, provides prevention methods.
5. Diabetes and Me	• Two domains: Informative and Interactive • Informative domain includes information, guidance to control diabetes. • Interactive domain records data of cholesterol level, blood sugar reading, BMI etc.

On the other hand some studies have been carried out for analyzing the risk and calculating probabilities of developing different diseases. Chawla and Davis [22] have proposed a patient-centered framework using Big Data driven approach along with collaborative filtering.Their study concentrates on finding disease risk factors of heart diseases, hepatitis, cancer and few more. Another study [23] lays its spotlight on diabetes, considering factors like HbA1c, age, sex, hypertension and uses data mining technique. An attempt was made to diagnose diabetes based on patient's habitual data and A1c sugar level in [24]. Marx in his study [25] found Gene factors, age, excess weight gain, unhealthy diet and lifestyle, environmental pollution as common factors that contribute in the development of type 2 diabetes.

Based on this literature review and to the best of our knowledge we have found that: a. only a few diabetes related mHealth applications have been developed in context of Bangladesh, b. very few studies have been conducted to propose prediction systems for diabetic patients c. no study has been found that explicitly revealed the factors of being affected by diabetes in context of Bangladesh. Therefore, this work focuses on developing an intelligent prediction application based on machine learning technique which will tell the users about their (users) probability of becoming a diabetic patient.

III. Research Methodology

From methodological point of view, this study has used the survey method to collect the empirical data. To attain the objectives of this work, the survey questionnaires were related to people's age, Body Mass Index (BMI), gender, HbA1c sugar level and ancestral diabetic history. Around 191 usable responses were received during May to July, 2017. Among the total 191 respondents 36.8% were female and 63.2% were male with average age of 36 ± 15 (m $\pm$ sd) and an average weight of 64 ± 16 (m $\pm$ sd). Though 68.5% of the respondents were unaware of their sugar level (HbA1c value), but the respondents who were diabetic patients were well aware of their sugar level. The data collected through survey have been analyzed using the Pearson Correlation and χ^2 test and the factors were found to be independent. So, the Naive Bayes [26–28] technique has been used to implement the intelligent diabetes prediction application.

IV. Implementation of the Diabetes Prediction Application

A. The Prediction Algorithm

For implementing the diabetes prediction application, an algorithm has been developed based on the Naive Bayes Classifier to classify the users as diabetic, prediabetic or nondiabetic. Four mutually independent features of patients used in our work are: a) age, b) BMI, c) gender, and d) ancestral diabetic history.

In Naive Bayes [27], if C is the estimated class and $\{x_1, x_2, ..., x_n\}$ are the features, the joint probability is defined as

$$P(C|x_1, x_2, ..., x_n) = P(C)\Pi_{i=1}^{n} P(x_i|C) \tag{1}$$

Here, $P(C|x_1, x_2, ..., x_n)$ represents the probability of being in class C having features $\{x_1, x_2, ..., x_n\}$. The prediction algorithm [Algorithm 1] contains two phases: training phase and testing phase.

1) Training Phase: The first phase of this algorithm is for training the system [line number 3-7 in Algorithm 1]. The steps followed in this phase are:

- The system begins by taking the related training data on features and classes (diabetic, prediabetic, nondiabetic) as input and enhances its knowledge base.
- Then, the mean (μ) and variance (σ) for age and BMI have been calculated for each class.
- Next, the conditional probability for categorical test data, gender and ancestral diabetic history is calculated using equation 2 [28], where f represents categorial value for feature x (e.g., 'male' is a categorial value of feature 'gender') then for class C_i.

$$P(x|C_i) = \frac{k}{\text{No. of training data in } C_i} \tag{2}$$

Algorithm 1 Predicting the Possibility of Diabetes

1: $features \leftarrow \{age, BMI, gender, inheritance\}$
2: $classes \leftarrow \{diabetic, prediabetic, nondiabetic\}$
3: **for** each c_i in classes **do**
4: calculate μ and σ of *age* and *BMI*
5: **end for**
6: calculate $P(gender|c)$ where $gender \in \{male, female\}$ and $c \in classes$
7: calculate $P(inheritance|c)$ where $inheritance \in \{yes, no\}$ and $c \in classes$
8: **for** each case in test data **do**
9: **for** each c_i in classes **do**
10: $posterior(c_i) \leftarrow \frac{P(c_i)\Pi P(x_i|c_i)}{evidence}$ where $x_i \in features$
11: **end for**
12: $max\_posterior \leftarrow max(posterior(c_i \in classes))$
13: $resulting\_class \leftarrow c_i$ such that $c_i \in classes$ and $posterior(c_i) = max\_posterior$
14: **end for**

where, $k =$ No. of occurrence of f for feature x in training data of C_i.

2) Testing Phase: The second phase is for testing the system [line 8-14 in Algorithm 1]. The steps are given below:

- For test data, the conditional probability for age and BMI $P(x|C)$ are calculated using equation 3, since these factors are assumed to exhibit normal (Gaussian) distribution in the data set.

$$P(x|C) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (3)$$

- For each set of test case, their posterior probability $P(C|x)$ of belonging to each predefined class is calculated using equation no 1. The division by evidence is intentionally ignored as it scales the result for all the classes by same value.
- After that, the class having maximum posterior value is taken to be the resultant class for that particular data set using the MAP (maximum a posteriori) decision rule.

B. Evaluating the System

The outcome of this analysis suggest non-diagnosed persons on possible diabetes threat using three categories:

- Out of danger
- Prediabetic stage
- In danger

To estimate the error of the system, we used the *leaving-one-out-technique*. Firstly, we used the first sample as test data and the rest $n-1$ samples are used as training data. We repeated the same process $n-1$ times and calculated error for n samples using equation 4,

$$\hat{P}(\varepsilon) = \Sigma_i^n \hat{P}_i(\varepsilon|x) \quad (4)$$

Thus the resultant estimated error is, $\hat{P}(\varepsilon) = 0.358$.

C. Developing the mHealth Application

The prediction algorithm has been implemented in JAVA using the Intellij IDE and an application named 'Diabetes Predictor' has been built for this system using Android Studio. Then the predictive algorithm has been integrated with the mHealth app as its backbone. As shown in Figure 1, one has to give age, BMI, gender, diabetes inheritance history as input and the app will provide a result along with a suggestion in both English and Bengali language.

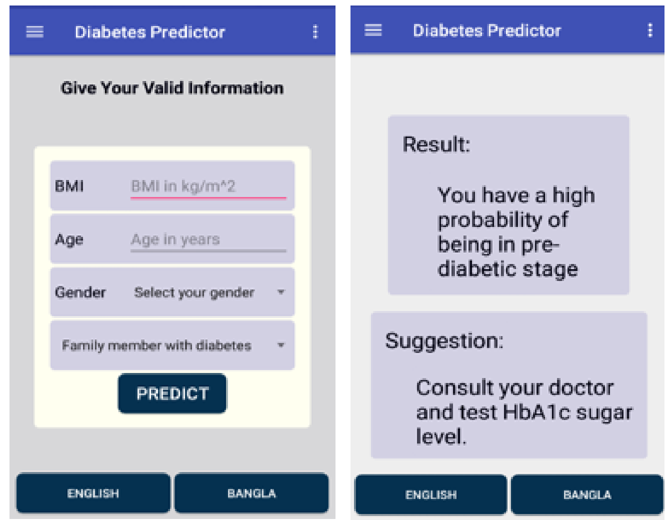

Fig. 1: Developed Intelligent mHealth Application

V. Discussion and Conclusion

A. Main Outcomes

In our study, data obtained have served as knowledge base for the development of the system. Finally, a software application is developed using basic features like age, BMI, gender, inheritance as these are known to all and users can easily provide the input to receive their results.

B. Practical Implications

Diabetes is a silent disease and people do not tend to visit hospitals on regular basis for diabetes tests. The intelligent prediction system that is available in the form of a software application helps any individual to know their probability of being diabetic sitting at home, reducing the efforts required to meet a physician in person. Being a software application the systems is cost effective, yet gives the result right away and provides users with enough time to prevent and control diabetes by making them aware of their present condition.

C. Study Limitation and Future Expansion

The primary limitation of the study is the lack of sufficiently large data set which eventually lowered the accuracy to 64% as data set constructs the knowledge base of the system which is utilized in learning stage. Again for making the application usable for general mass, only some commonly known basic features have been used which are not enough for correct prediction. Another limitation is the usability issues that are not considered explicitly to develop the mHealth system.

Our main focus for future expansion is the enhancement of knowledge base by collecting sufficiently large amount of data that will improve the accuracy of the system to a reliable extent. We also plan to include features like daily exercise, place of living, dietary habits, smoking habit, drinking habit as they have been found to be risk factors of diabetes in [29] and other studies also. Medical factors like hypertension, blood pressure, high cholesterol, polycystic ovary syndrome (for female) will also be considered in the predictive analysis of diabetes in our further research work as per the suggestion of some diabetes specialists.It is also expected that a feature will be included in the succeeding system which will provide medical advice to the ones who are under the threat of diabetes. Finally, an extensive usability evaluation study will be caved out to evaluate the usability and user experience (UX) of the application. Moreover, it has been found in several studies including [30] that the accuracy of Artificial Neural Network is considerably better than that of Naive Bayes classifier in case of diabetes prediction. Therefore we intend to implement the prediction system in Artificial Neural Network during further research.

D. Concluding Remarks

In this paper, survey method and machine learning techniques have been used for developing a predictive system that comes as a mobile application with a view to supporting diabetic patients. Considering the present context of explosion rate of diabetes across the world, and the vast use of smart phones in every sphere of life, it is expected that the proposed system will play a vital role in creating awareness among people about diabetes as well as reducing the outspread of diabetes. This study thus will contribute to provide effective health services as well as to create awareness about diabetes.

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